Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/23/2013
PI Name:	Pluth, Janice Ph.D.		
Project Title:	Elucidating the Relationship Between the Effects of Various Radiation Qualities and Cancer Development Processes using Novel Flow-based Assays		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	89154-9900	**Congressional District:**	1
Comments:	NOTE: Formerly at Lawrence Berkeley National Laboratory until fall 2017.		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	09/08/2008	**End Date:**	12/31/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2013, per PI and LBL info (Ed., 9/27/13) NOTE: Project will continue as new award; end date is now 9/30/2013, per PI and LBL info (Ed., 7/24/13) NOTE: Extended to 12/31/2014 per PI and LBL info (Ed., 7/12/13) NOTE: Extended to 7/1/2013 per PI and NASA info (Ed., 1/8/2013) NOTE: Extended to 12/31/2012, per C. Guidry/JSC (Ed., 6/11/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ08HD70I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current radiation quality factors used for determining cancer risk for space radiation do not provide the accuracy needed, and there is debate as to the relevance of the experimental studies they're based upon. Therefore, one of NASA's primary goals is to improve the understanding of the carcinogenic process related to high LET exposures to better define the risk. Recently our current understanding of carcinogenesis has improved, and a number of cancer hallmarks have been identified. Some hallmarks, or surrogate cancer endpoints include genetic instability, telomere length changes, and aberrant cellular signaling. Identifying relationships between early and late cell signaling profiles, later surrogate cancer endpoints, and radiation qualities will aid in determining a better risk assessment for cancer formation. In addition, using relevant cell types, such as human epithelial cells in which the majority of cancers develop is also key. In this proposal we will study critical phosphorylation signaling at early and late time points as well as surrogate cancer endpoints of telomere length changes and genomic stability as measured by centrosome amplification, at these same late times post exposure. Epithelial and fibroblast cells from the same individuals will be used to better understand the etiology of cancer, and why certain cell types more easily become carcinogenic. We will also perform studies to understand the heterogeneous nature of high LET radiation. Energy deposition from high LET ions is dense in the track cores, resulting in closely spaced and complex damages, and delta rays originating from cores provide additional exposures similar to X-rays. Using carefully chosen charge, energy and LETs of particles to obtain differing populations of cells hit by tracks and delta rays alone we will investigate how phosphorylation profiles change. These studies will allow us to better quantify the effects of space radiation and its relationship to the carcinogenic process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	In this proposal we have been focusing on defining the kinetics for critical phosphorylation signaling at early (< 24 h) and late time points (days-weeks) following high and low LET radiation exposure. In addition, we have evaluated how radiation exposure affects telomere length and centrosome aberrations, as noted markers of carcinogenic risk. Breast cancer studies have revealed that telomere length decreases during progression from usual ductal hyperplasia (UDH) to invasive cancer, and centrosome amplification has been detected in a broad range of solid tumors and in leukemia and lymphoma. Additionally, centrosome abnormalities have been shown to increase with increasing tumor grade and aggressiveness. Both of these potential biomarkers for carcinogenesis are being quantified using flow cytometry-based assays and compared to phospho-signaling kinetics following radiation exposures. Pairs of mammary epithelial and fibroblast cells from the same individuals have been used to better understand the etiology of cancer, and whether certain cell types respond differentially to radiation exposure. These studies complement our earlier studies using a normal hTERT immortalized skin fibroblast line, and help us to identify unique signaling patterns specific to different cell types. Track structure effects have been investigated by carefully choosing particles of various energies to obtain differing populations of cells hit to a greater or lesser extent by the core particle or delta-rays to then investigate how phosphorylation profiles and surrogate cancer endpoints are affected by these radiation quality components. The intent of this proposal is to provide a comprehensive picture of HZE effects from early to late phosphorylation profiles, and the relationship to various surrogate markers for carcinogenesis. Materials & Methods: Cells were set up in flasks or chamber slides prior to irradiation, irradiated at the various doses and then fixed at multiple time points thereafter. A methanol and a combination paraformaldehyde and methanol fixation were used. Fixed slides were sent back from Brookhaven to our home institution in PBS to stain for the appropriate primary and secondary antibodies. Cells exposed in flasks were trypsinized and fixed at various times after exposure, and then sent back on ice in methanol. Aliquots of fixed cells were then stained for various phospho-proteins and analyzed via flow cytometry. For analysis at longer timepoints after exposure (weeks) cells were grown to confluence while at Brookhaven then frozen down before leaving (~ 1-2 days post exposure). Cells were then thawed back at the home institution, and expanded to then analyze for telomere length changes and centrosome defects. Cells were fixed with a methanol fix for flow cytometry based analysis of centrosome dysfunction, and were hybridized to a telomere probe to allow detection of telomere length changes. Brief summary of past years research accomplishments: Normal hTERT immortalized fibroblasts were exposed to a very comprehensive range of carefully chosen high LET ions and energies (Fe 300, 600, and 1000 MeV/u; Ti 300, 326, and 386 MeV/u; Si 93, 300 and 1000 MeV/u at timepoints between 0.5 and 24 h), to identify the relative contribution of delta rays and track structure effects based on the phospho-kinetics of gamma H2AX, pSMC1, and pATF2. These studies revealed Z2/beta2 to be a primary determinant in the phosphorylation response for gamma H2AX and pATF2, however unique radiation quality dependent cell cycle blocks appeared to play a large role in pSMC1 kinetics. Cell cycle kinetic data accessed from the data thus added an additional dimension to our understanding of the response of cells to the various radiation qualities. Linear regression analysis has also been performed and the total of this data has been used to draft into two manuscripts that will be submitted in the upcoming month. To determine whether the phospho-kinetic profiles defined for the normal hTERT immortalized fibroblast line were unique to this cell line, we also compared phosphorylation profiles to the profiles observed in primary mammary fibroblasts and genetically matched mammary epithelial cells following a subset of the radiation exposures. Thus far we have compiled median levels of fluorescence for the primary mammary fibroblast and isogenic epithelial cells and exported data to JSC to have individual cell data analyzed once controls have been normalized as was completed for the normal fibroblast skin cells. Normalization may change results significantly, thus we are unable to make any absolute conclusions at this point however studies suggest that primary cells, as opposed to hTERT immortalized cells reveal much dampened levels of phospho-protein induction following damage. Cell type and phospho-protein dependent differences were also noted in terms of the degree of sensitivity to various radiation qualities, but conclusions will be dependent upon normalization results. In the process of performing this analysis we have had some additional difficulties with primary epithelial and fibroblast cells not noted with hTERT skin fibroblasts that we've needed to address. In both primary skin fibroblasts and epithelial cells two defined populations based on forward scatter (size) and side scatter (internal complexity) are noted via flow cytometry analysis. We have needed to separately analyze both of these populations due to differences in phospho-signal levels and this has added additional complexity to this analysis. The proportion of each of these populations changes with radiation and time following radiation. We are currently also completing further studies characterizing these populations in terms of senescence markers and chromatin changes. Conclusions from all phospho-kinetic studies thus far include that 1) high LET induces a greater induction of residual signaling at 24 h as compared to equal doses of low LET; and 2) that at early timepoints ATM-mediated phospho-inductions are much lower following all high LET ion exposures as compared to low LET, indicating that the ATM protein is hampered in fully activating down stream proteins following high LET irradiation as compared to low

LET exposures, even at higher high LET doses where all cells in the population should be hit.

In order to better understand the relationship between radiation exposure and carcinogenesis we have analyzed how these various radiation qualities effects two surrogate cancer endpoints, telomere length and centrosome amplification. Flow cytometry-based telomere length assay, or flow FISH was used to determine how various qualities of radiation affected telomere lengths of cells differentially. In the past year additional data was obtained for all radiation qualities so that three independent sets of data are now available for all radiation qualities. In addition data has been exported to JSC to analyze on single cell basis how radiation quality affects the length of telomeres in populations of cells within each exposed group. Due to high LETs nature of exposing cells to unequal levels of energy deposition it is expected that proportions of the cells may show greater or lesser effects. Thus various radiation qualities may affect the telomere length of the sub-populations of cells exposed differentially. Comparing the median telomere length changes a dose-dependent shortening of telomeres for both mammary epithelial and fibroblast cells following high LET exposures was observed, whereas low LET exposure resulted in only a small similar decrease following all doses up through 1Gy. Mammary epithelial cells were revealed to be more sensitive to radiation-induced telomere shortening than fibroblasts. In comparing telomere length changes based on fluence unique differences were noted between fibroblasts and epithelial cells as well as between radiation qualities. Fibroblasts in general showed the greatest resistance to radiation exposure, although Fe 300 MeV/u revealed a very similar response between the cell types. This may reflect the greater effectiveness of Fe 300 with the greatest delta ray deposition. Fe 1000 was also unique in being the only radiation quality in which fibroblasts showed a higher sensitivity.

Additional work has been needed during this last year to aid in optimizing the centrosome flow-based assay. Centrosome analysis on various days after exposure have been preformed as well as microscopic immunofluorescence imaging done side-by-side. These studies will aid in determining the induction time and how to best determine flow cytometry assay cut offs based on results from microscopic scoring of centrosome aberrations on slides. Our collaborators at JSC now have all of the exported data for these optimizations and experiments for all radiation qualities. Pending optimization results we will then analyze the complete set of centrosome studies for the radiation qualities which telomere analysis has been completed. Preliminary data suggest a greater level of centrosome aberrations with dose, but no clear evidence of a dose response. Also in preliminary centrosome defect comparisons between fibroblast and epithelial cells following X-ray exposure we have noted that only epithelial cells show an increase in centrosome defects with radiation exposure. This may relate to the increased risk of epithelial cells to form tumors following carcinogenic exposures. Further studies will be done comparing centrosome defects in these cell types after high LET radiation exposure. We have also looked at phospho-signaling in these cells which have been grown for weeks post exposure and see evidence of residual foci many weeks post exposure, in some cases even residual tracks appear evident in cells, perhaps indicating delayed genomic instability and possibly increased levels of ROS in these previously irradiated cells. We are currently investigating how ROS levels may vary in cells after previous exposure to high or low LET and whether differences are noted based on cell types.

In summary the telomere, centrosome, and phospho-kinetic studies reveal that high LET is more effective than low LET in inducing high levels of these endpoints, and that fibroblasts appear more resistant than epithelial cells. Studies for centrosome defects are still incomplete however evidence to date seems to suggest a lack of a dose response, and that fibroblasts show a greater resistance as compared to epithelial cells. Phospho-signaling results for primary mammary epithelial and fibroblast cells still need to be analyzed following normalization and are currently underway.

Immunofluorescence studies completed on the same cells in which flow cytometry assays have been performed reveal phospho-signaling is evident in cells many weeks post exposure and that there is an increase in phospho-signaling in cells that also have centrosome defects, linking phospho-signaling for damage response proteins to this indicator of genomic instability In the upcoming year we will complete the analysis for the data we have already collected and compile data for at least two papers centering on telomere length changes following radiation and its relationship to phospho-signaling and a second on centrosome aberrations in relation to phospho-signaling and radiation quality. This data analyzed on a single cell basis following numerous radiation quality exposures should aid in supporting efforts to further reduce the uncertainty of the carcinogenic risk from high LET exposures.

Bibliography Type: Description: (Last Updated: 11/01/2023)

Articles in Peer-reviewed Journals

Wang M, Saha J, Hada M, Anderson JA, Pluth JM, O'Neill P, Cucinotta FA. "Novel Smad proteins localize to IR-induced double-strand breaks: interplay between TGFß and ATM pathways." Nucleic Acids Res. 2013 Jan;41(2):933-42. http://dx.doi.org/10.1093/nar/gks1038 ; PubMed PMID: 23221633 , Jan-2013